Group13
Players Profile and Gaming Platform

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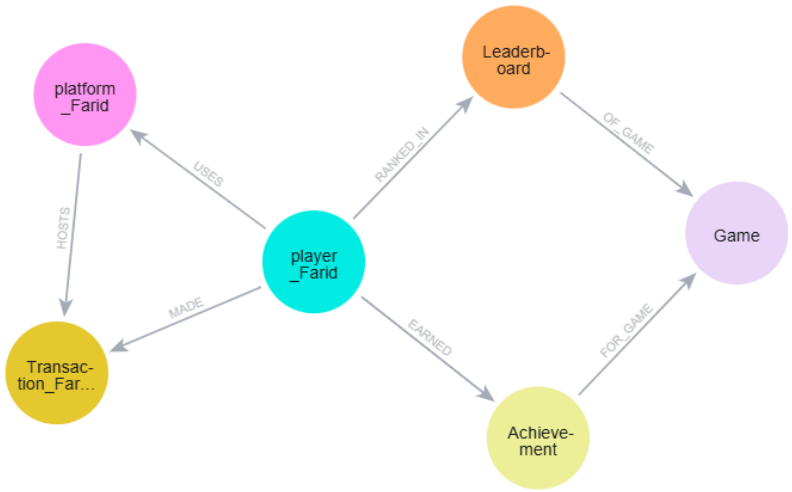
Contributions

Farid Ahmad Tokhi – Platforms, Transactions and Players

Alif Hossain – Achievements, Leaderboards, Game

Schema


platform
_Farid
Leaderb-
oard
USES
RANKED_IN
OF_GAME
HOSTS
player
_Farid
Game
MADE
EARNED
FOR_GAME
Transac-
tion_Far...
Achieve-
ment

SIMPLE QUERIES

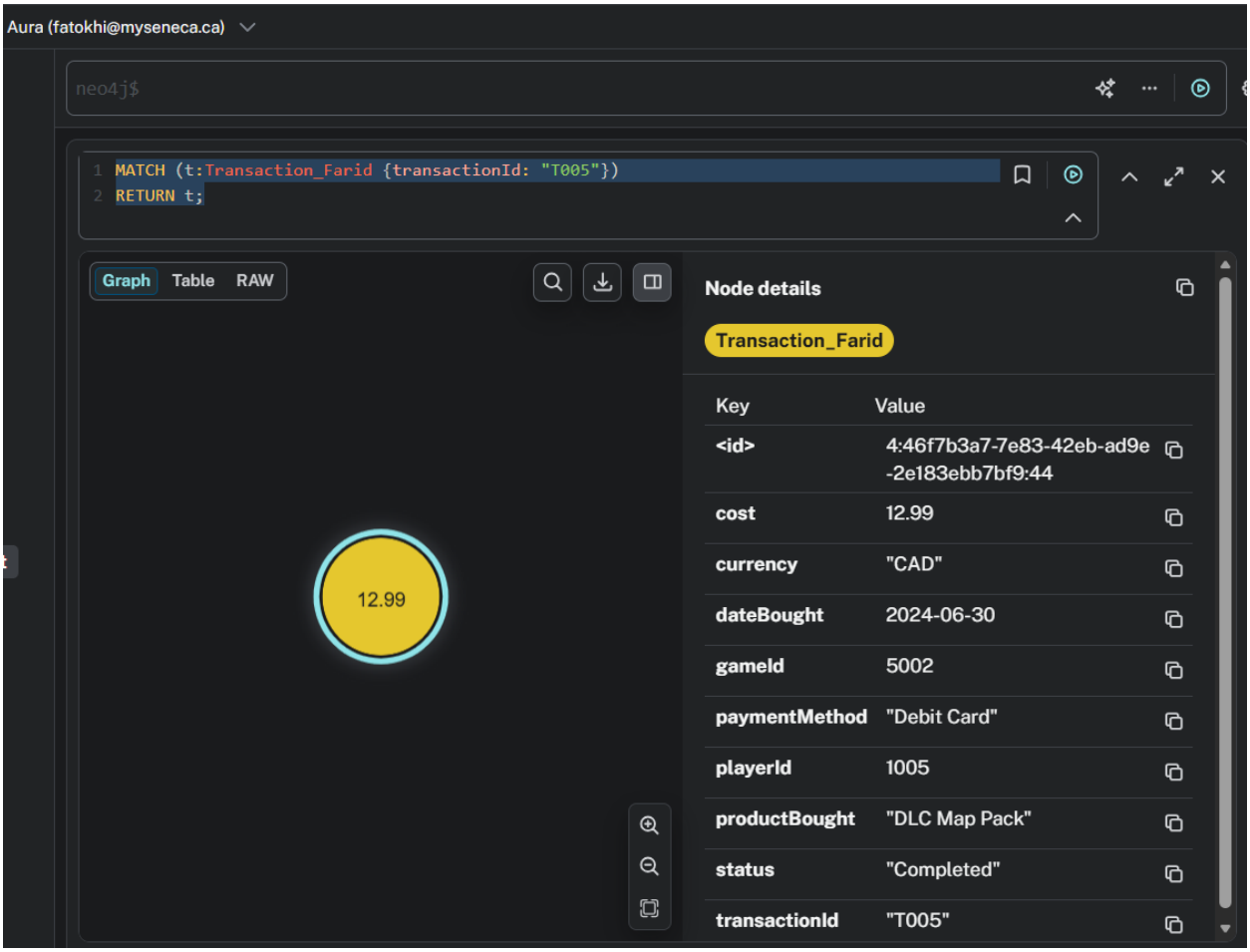
Simple Query 1 (Farid Ahmad Tokhi):

```
MATCH (t:Transaction_Farid {transactionId: "T005"})
RETURN t;
```

Description:

This query returns the details of a single transaction, identified by its unique `transactionId`.
It show how to retrieve one specific record using filtering.

Output:



Simple Query 2 (Farid Ahmad Tokhi):

```
MATCH (pl:player_Farid)-[:USES]->(pf:platform_Farid)
WHERE pf.platformName = "PlayStation Network"
RETURN pl.playerId, pl.username, pf.platformName;
```

Description:

This query finds all players who are connected to the PlayStation Network platform through the USES relationship.

Output:

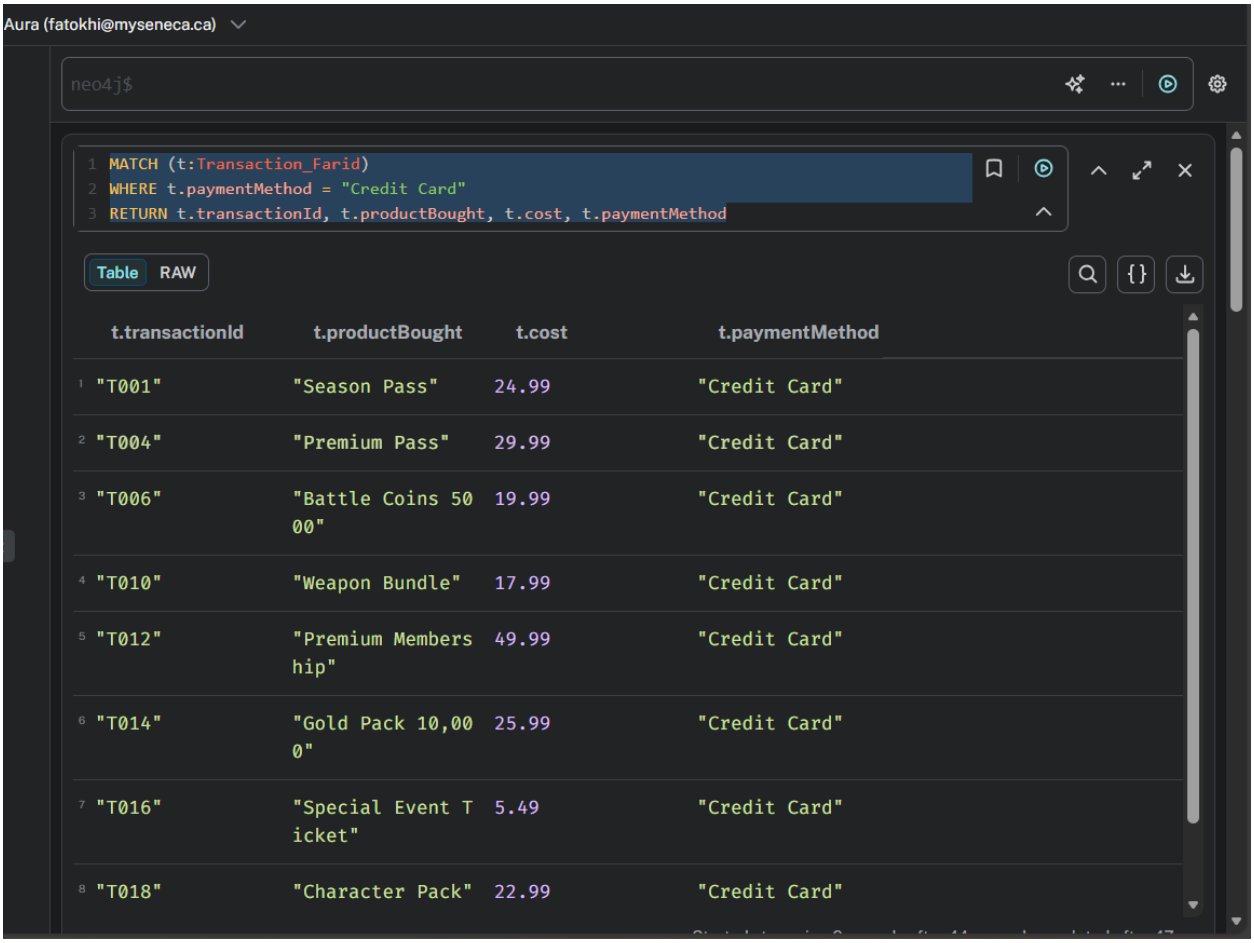
Simple Query 3 (Farid Ahmad Tokhi):

```
MATCH (t:Transaction_Farid)
WHERE t.paymentMethod = "Credit Card"
RETURN t.transactionId, t.productBought, t.cost, t.paymentMethod
```

Description:

This query filters all transactions to return only those where the payment method is Credit Card.

Output:



	t.transactionId	t.productBought	t.cost	t.paymentMethod
1	"T001"	"Season Pass"	24.99	"Credit Card"
2	"T004"	"Premium Pass"	29.99	"Credit Card"
3	"T006"	"Battle Coins 5000"	19.99	"Credit Card"
4	"T010"	"Weapon Bundle"	17.99	"Credit Card"
5	"T012"	"Premium Membership"	49.99	"Credit Card"
6	"T014"	"Gold Pack 10,000"	25.99	"Credit Card"
7	"T016"	"Special Event Ticket"	5.49	"Credit Card"
8	"T018"	"Character Pack"	22.99	"Credit Card"

ADVANCED QUERIES

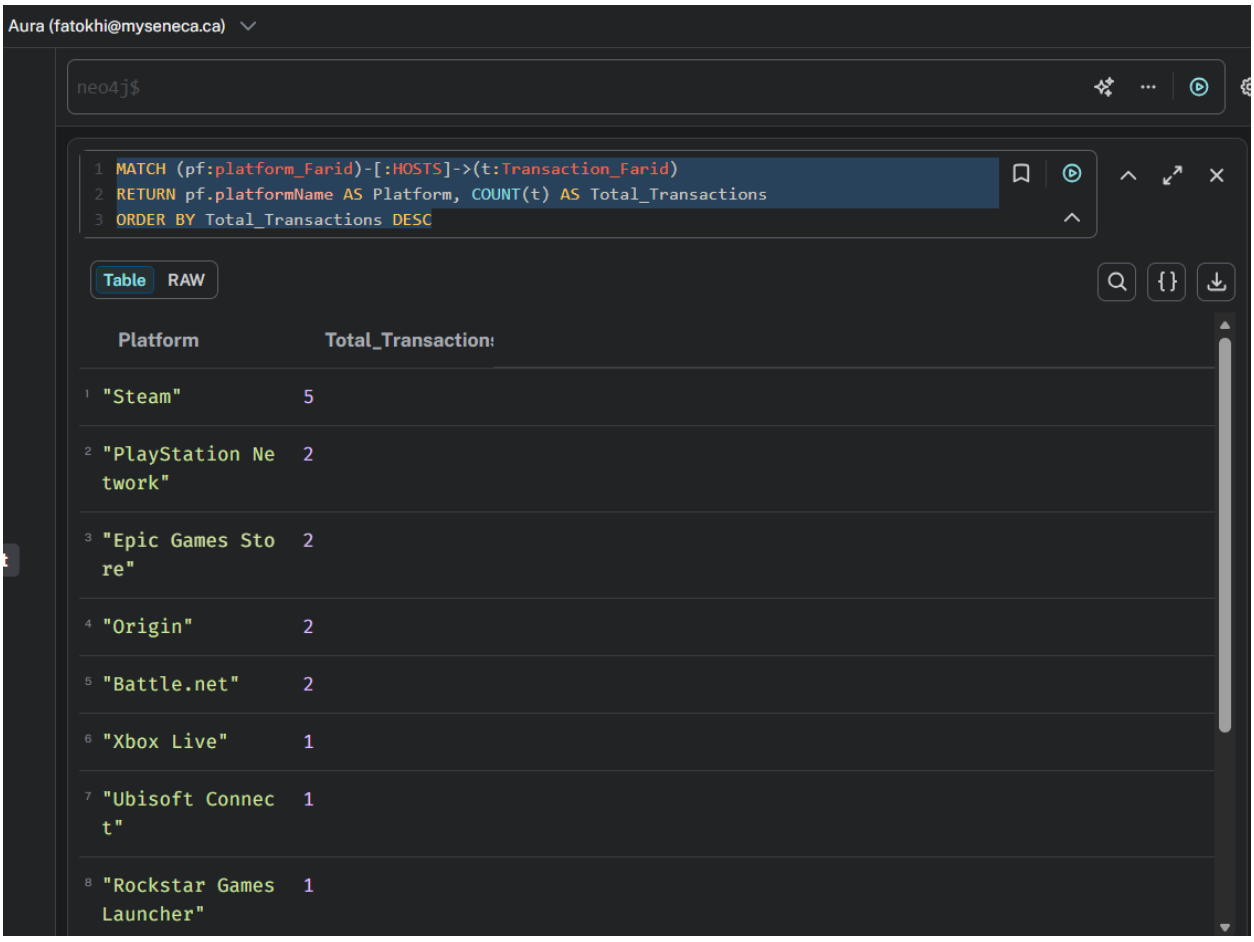
Advanced Query 1 (Farid Ahmad Tokhi):

```
MATCH (pf:platform_Farid)-[:HOSTS]->(t:Transaction_Farid)
RETURN pf.platformName AS Platform, COUNT(t) AS Total_Transactions
ORDER BY Total_Transactions DESC
```

Description:

This query aggregates all transactions hosted by each platform.
It uses pattern matching and the COUNT( ) function to summarize how many purchases were made on every platform.

Output:



Advanced Query 2 (Farid Ahmad Tokhi):

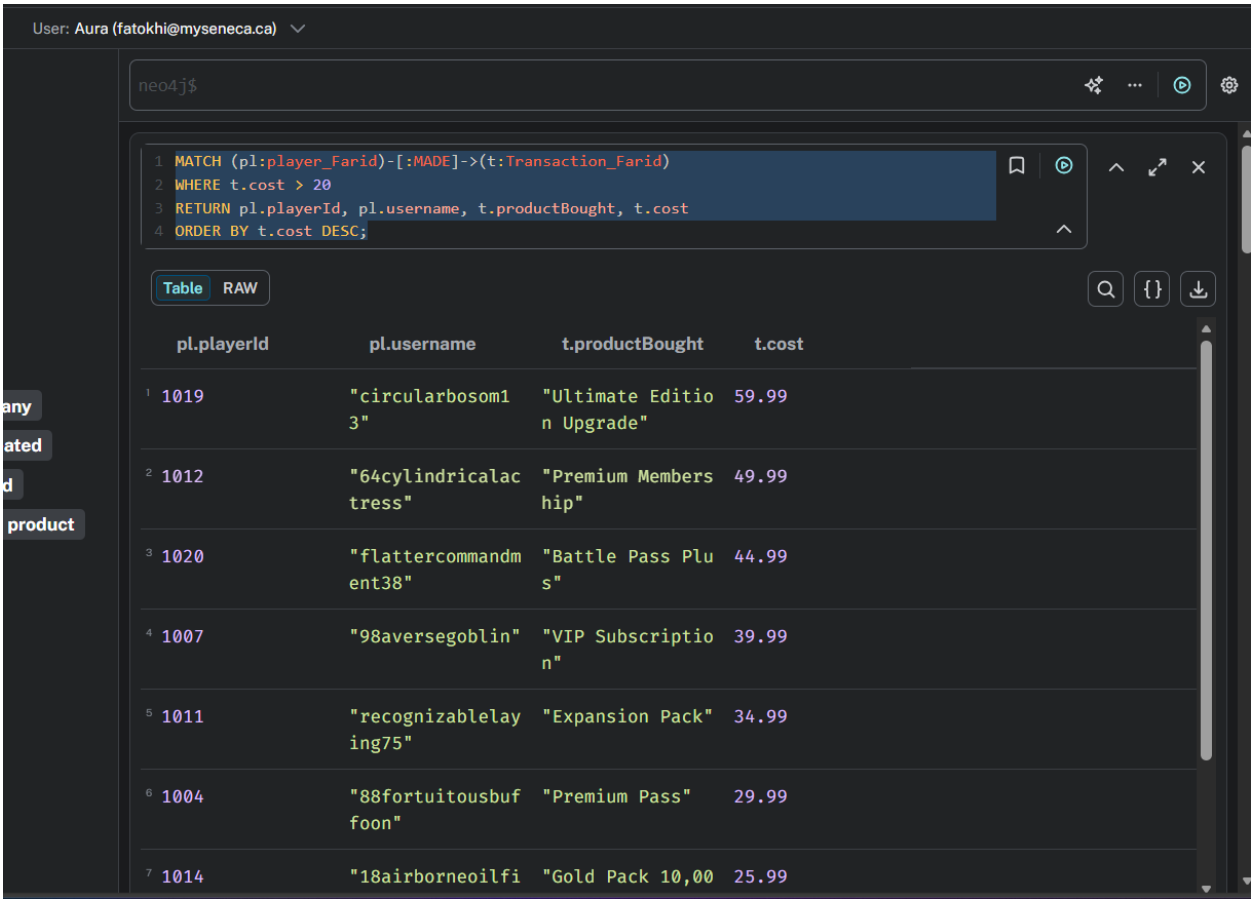
```
MATCH (pl:player_Farid)-[:MADE]->(t:Transaction_Farid)
```

```
WHERE t.cost > 20
RETURN pl.playerId, pl.username, t.productBought, t.cost
ORDER BY t.cost DESC;
```

Description:

This query filters the transactions to find players who made purchases costing more than $20.

Output:



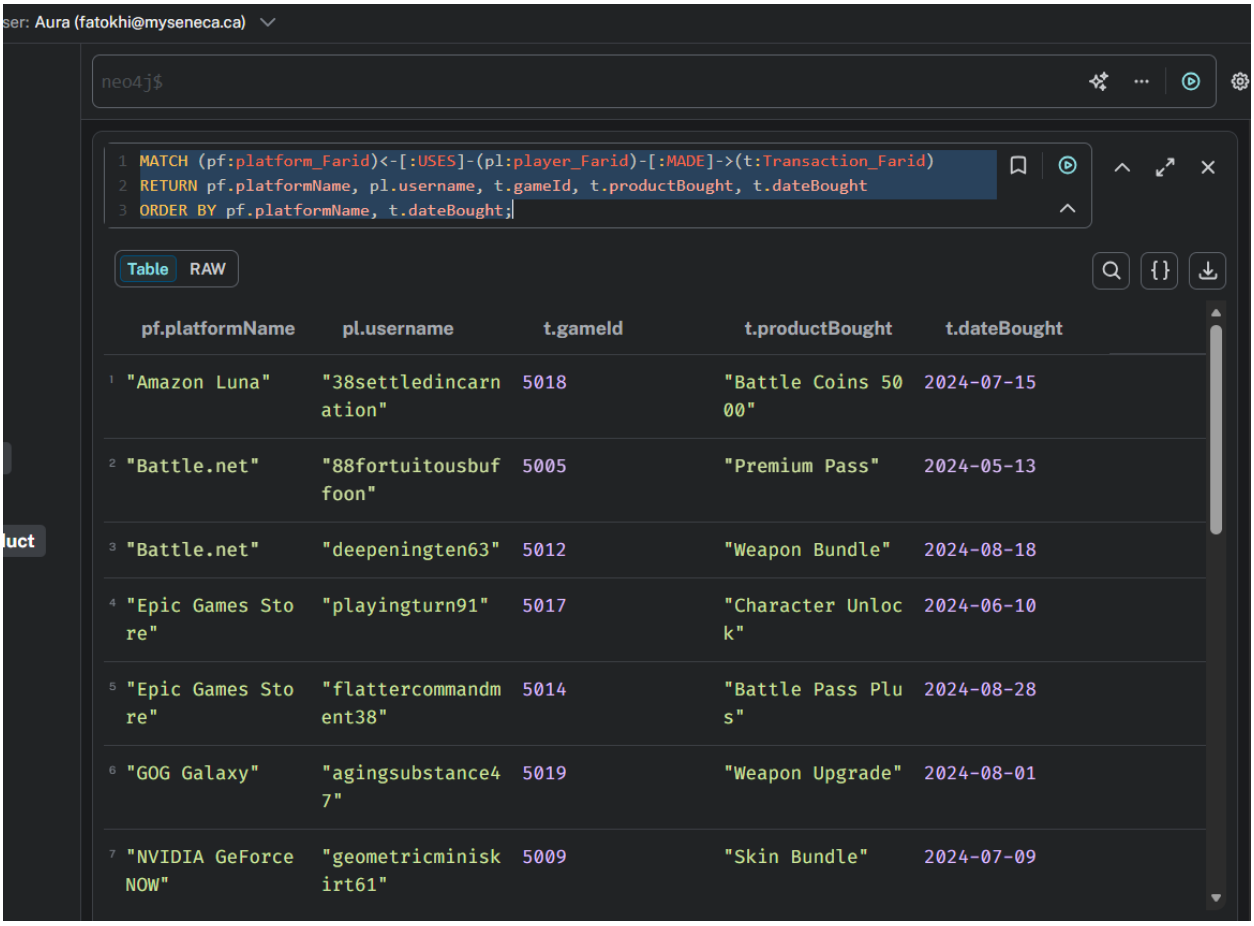
Advanced Query 3 (Farid Ahmad Tokhi):

```
MATCH (pf:platform_Farid)<-[:USES]-(pl:player_Farid)-[:MADE]->(t:Transaction_Farid)
RETURN pf.platformName, pl.username, t.gameId, t.productBought, t.dateBought
ORDER BY pf.platformName, t.dateBought;
```

Description:

this traversal query combines data from three node types: platform, player, and transaction.
It shows which players on each platform made purchases, which game they relate to, and the time of purchase.

Output:



SIMPLE QUERIES (Alif)

Simple Query 1:

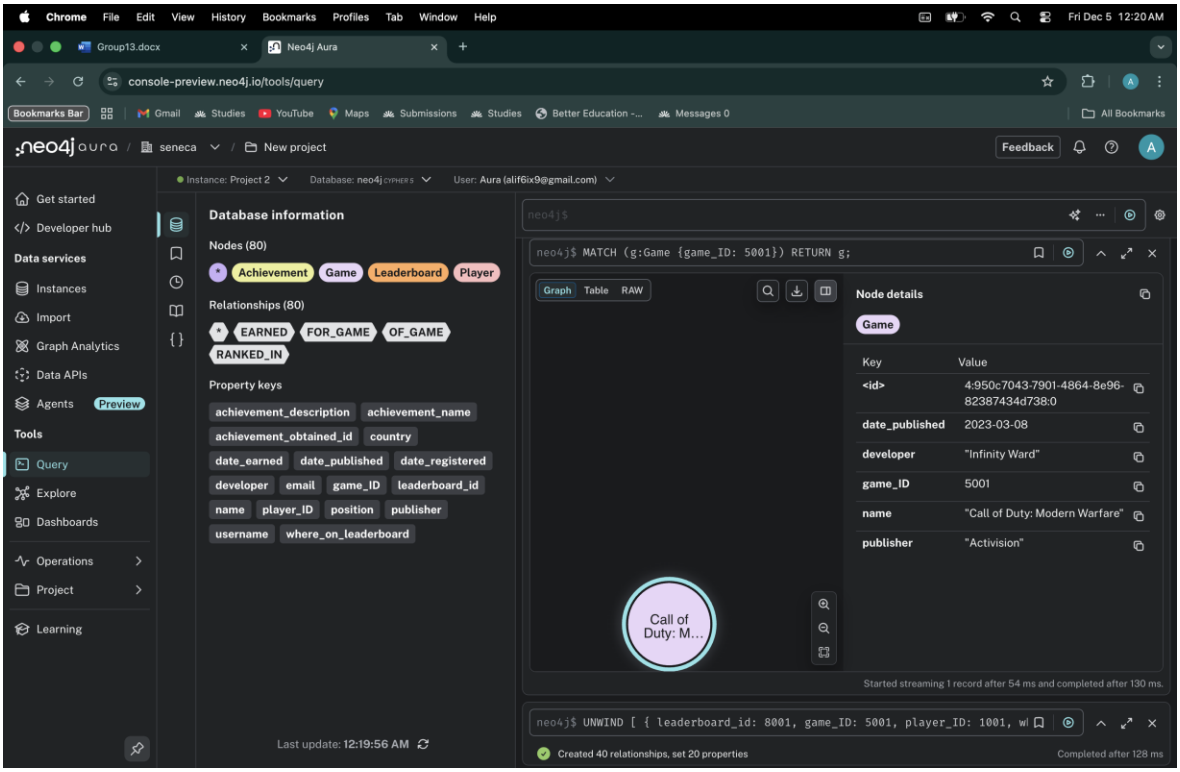
MATCH (g:Game)

```
RETURN g.game_ID, g.name, g.developer, g.publisher, g.date_published
LIMIT 10;
```

Description:

This query returns the details of a single game, identified by its unique game_ID.
It demonstrates how to retrieve one specific record from the Game node label by applying a simple equality filter.

Output:



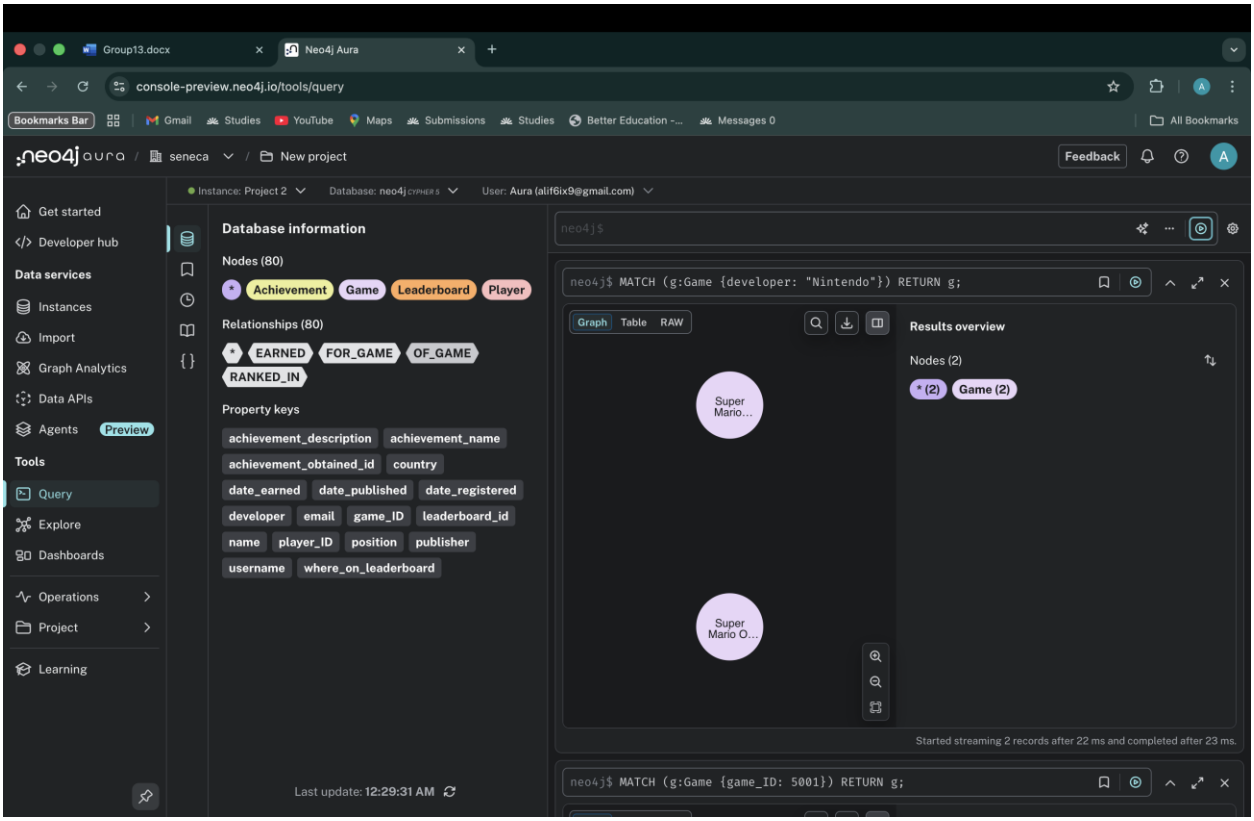
Simple Query 2:

```
MATCH (g:Game {developer: "Nintendo"})
RETURN g;
```

Description:

This query retrieves all games created by a specific developer, in this case Nintendo.
It demonstrates how to filter nodes based on a non-unique attribute to return multiple matching records.

Output:



Simple Query 3:

MATCH (p:Player {player_ID: 1001})-[:EARNED]->(a:Achievement)

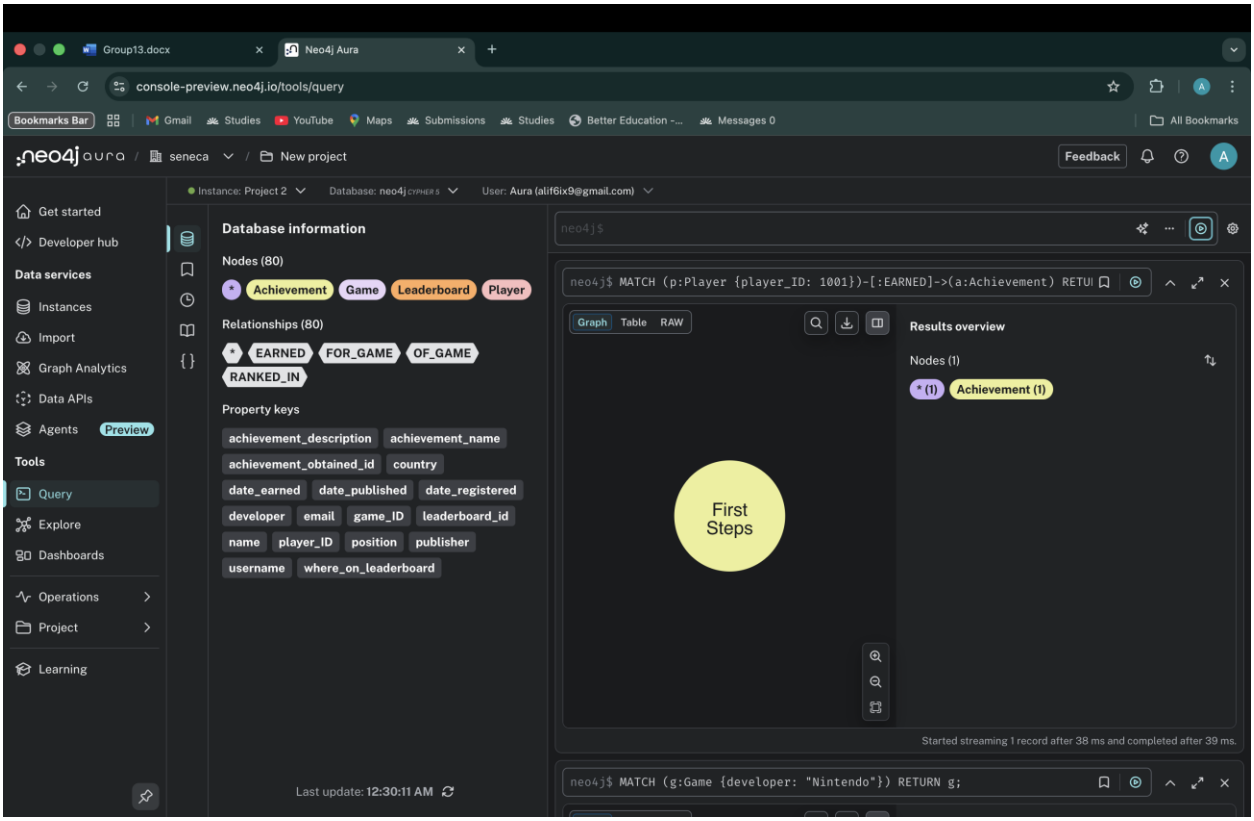
RETURN a;

Description:

This query returns all achievements earned by a specific player, identified by their unique player_ID.

It demonstrates how to traverse a relationship (EARNED) to retrieve connected records from another node label.

Output:



ADVANCED QUERIES (Alif)

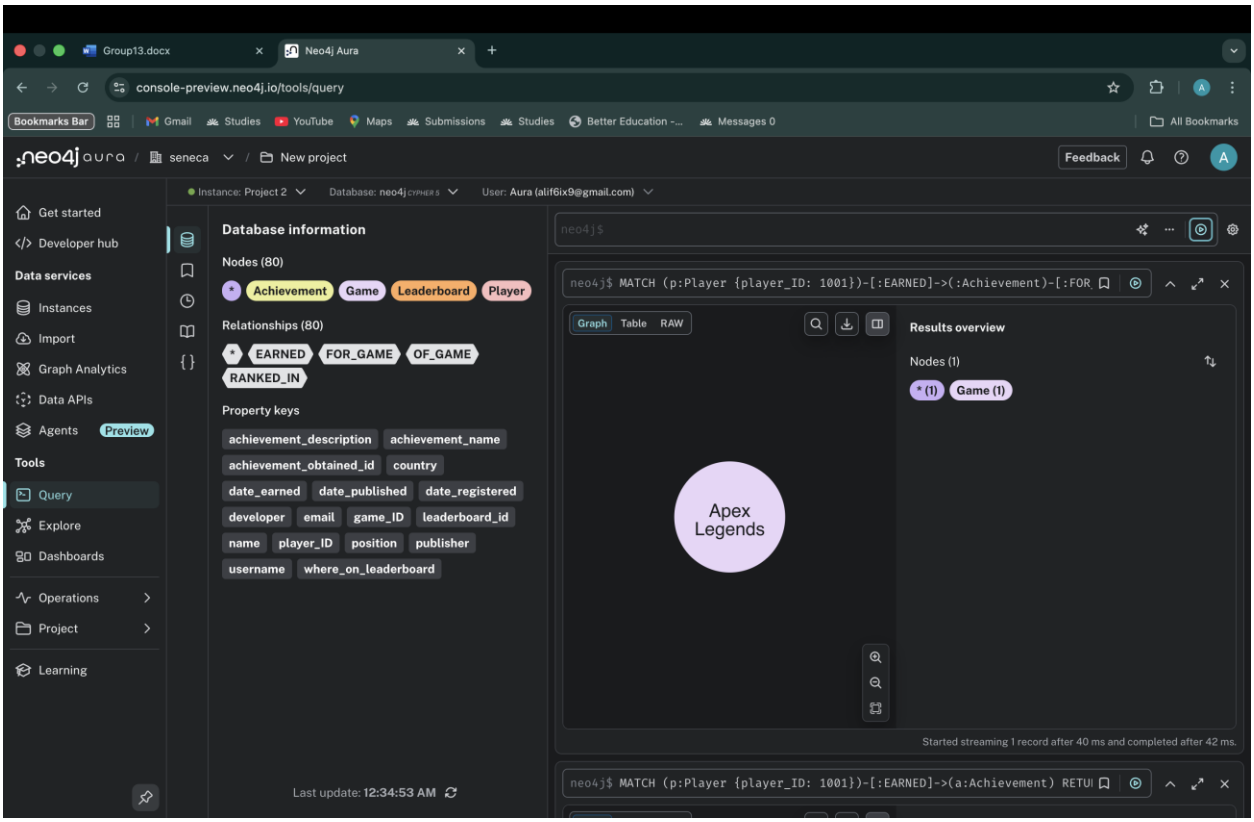
Advanced query 1:

MATCH (p:Player {player_ID: 1001})-[:EARNED]->(:Achievement)-[:FOR_GAME]->(g:Game) RETURN g;

Description:

This query performs a multi-hop traversal from a Player to their Achievements, and then to the Games where those achievements were earned.
It demonstrates pattern matching across multiple connected node types to retrieve related records.

Output:



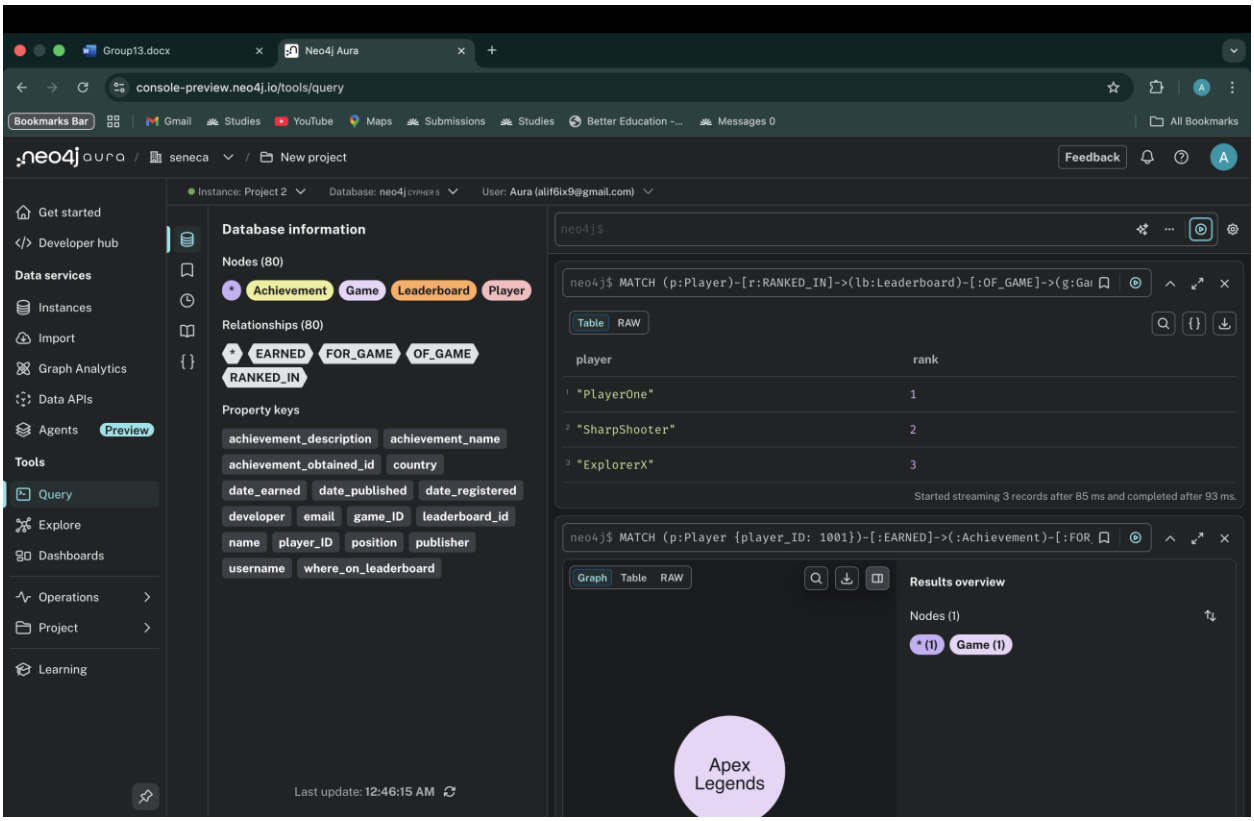
Advanced query 2:

MATCH (p:Player)-[r:RANKED_IN]->(lb:Leaderboard)-[:OF_GAME]->(g:Game {game_ID: 5001})
RETURN p.username AS player, r.position AS rank ORDER BY rank ASC LIMIT 3;

Description:

This query retrieves the top three players ranked on the leaderboard for a specific game (game_ID: 5001).
It demonstrates how to combine filtering, relationship traversal, sorting, and limiting to return ranked leaderboard data.

Output:



Advanced query 3:

MATCH (p:Player)-[:EARNED]->(a:Achievement)

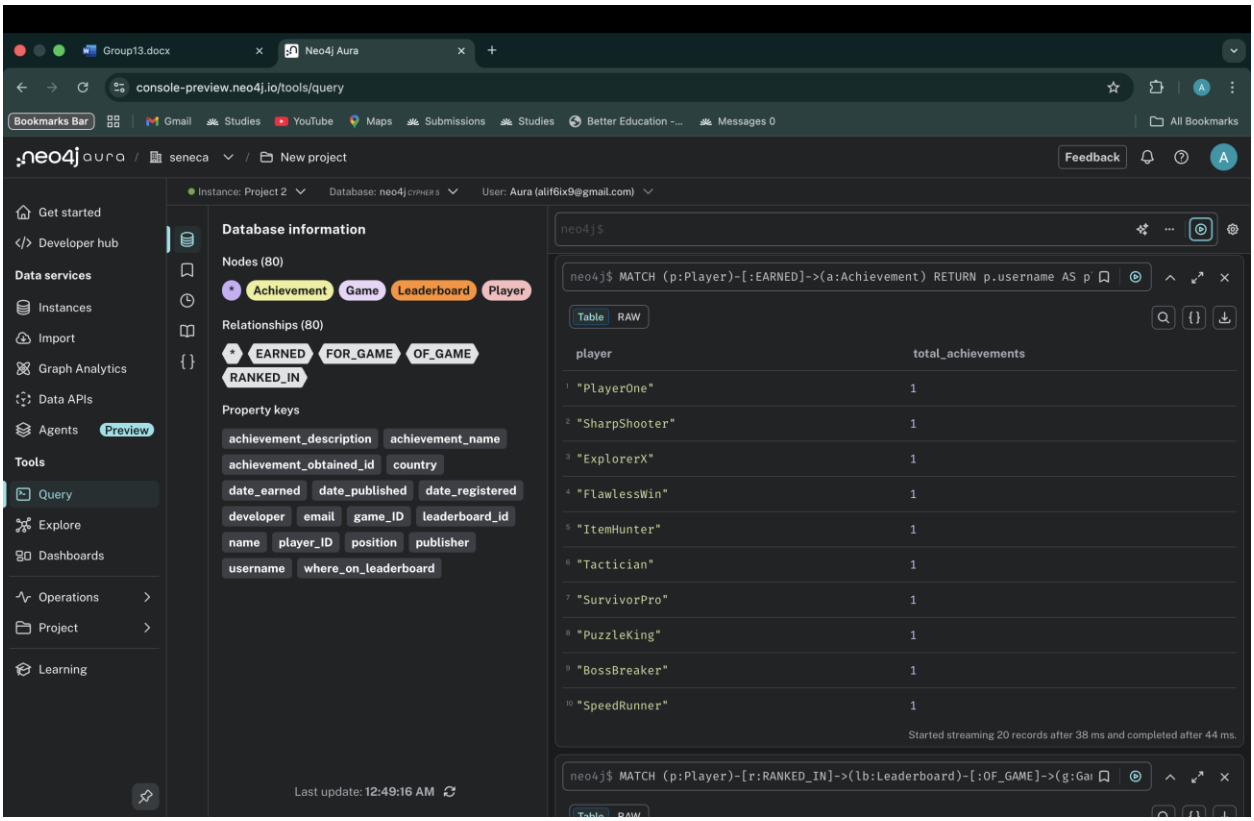
RETURN p.username AS player, COUNT(a) AS total_achievements

ORDER BY total_achievements DESC;

Description:

This query uses aggregation to calculate how many achievements each player has earned. It demonstrates grouping results by player and applying the COUNT() function to summarize connected achievement nodes.

Output:


Group13.docx
Neo4j Aura
console-preview.neo4j.io/tools/query
neo4j aura / seneca / New project
Feedback
Instance: Project 2
Database: neo4j
User: Aura (alif6ix9@gmail.com)
Get started
Developer hub
Data services
Instances
Import
Graph Analytics
Data APIs
Agents
Preview
Tools
Query
Explore
Dashboards
Operations
Project
Learning
Database information
Nodes (80)
Achievement
Game
Leaderboard
Player
Relationships (80)
EARNED
FOR_GAME
OF_GAME
RANKED_IN
Property keys
achievement_description
achievement_name
achievement_obtained_id
country
date_earned
date_published
date_registered
developer
email
game_ID
leaderboard_id
name
player_ID
position
publisher
username
where_on_leaderboard
Last update: 12:49:16 AM
neo4j$ MATCH (p:Player)-[:EARNED]->(a:Achievement) RETURN p.username AS p
Table
RAW
player
total_achievements
"PlayerOne" 1
"SharpShooter" 1
"ExplorerX" 1
"FlawlessWin" 1
"ItemHunter" 1
"Tactician" 1
"SurvivorPro" 1
"PuzzleKing" 1
"BossBreaker" 1
"SpeedRunner" 1
Started streaming 20 records after 38 ms and completed after 44 ms.
neo4j$ MATCH (p:Player)-[r:RANKED_IN]->(lb:Leaderboard)-[:OF_GAME]->(g:Ga